

EAIRLE

Embodied AI & Robot Learning Engineer Master Curriculum

Day 3 Revised Packet - July 7, 2026

Important correction: We are pausing LeetCode-style DSA for a few days. You do not need to solve hash-map or interview problems before learning the concepts. Today will build the foundation needed for DSA later: loops, lists/arrays, strings, dictionaries, and NumPy sensor arrays.

Day 3 Objective

Today is about strengthening programming fundamentals for AI and robotics. We will not chase LeetCode today. The goal is to understand arrays, strings, loops, simple dictionaries, and NumPy arrays through small robotics-flavoured programs.

Revised Block Plan

Block	Time	Focus	Main output
1	2 hours	Python loops, functions, and clean programs	4 small programs + notes
2	1.5 hours	DSA foundation without LeetCode	List/string/dictionary drills
3	1 hour	NumPy first touch: sensor arrays	numpy_sensor_arrays.py
4	45 min	C++ basics: I/O, loops, vectors orientation	cpp_basics_notes.md + 2 tiny programs
5	25 min	Git + journal	Commit + progress report

Block 1 - Python Loops and Functions

Primary resource: CS50P Week 2 - Loops, and CS50P Week 3 debugging style if needed. Use this only as the course source. Do not open random Python playlists today.

Official documentation: Python Tutorial, Control Flow Tools. Read only the sections on if statements, for statements, range(), break/continue, and defining functions.

Write these files in Day03/:

File	Requirement
countdown_timer.py	Input a number and print countdown to zero.
sensor_threshold_checker.py	Given a list of mock sensor readings, print which readings are safe or unsafe.
robot_command_menu.py	Use a looped menu: forward, left, right, stop, exit.
function_calculator.py	Write add, subtract, multiply, divide as separate functions.

Block 2 - DSA Foundation WITHOUT LeetCode

We are not doing LeetCode today. The goal is to understand the building blocks before solving named interview problems.

Concepts to learn today: list traversal, index, length, string traversal, counting, and dictionary basics. A dictionary in Python is the beginner version of what interview people call a hash map. We will learn it slowly through examples.

Create `dsa_foundation_drills.py` with these drills:

- Given a list of numbers, print each number with its index.
- Find the maximum and minimum number in a list without using `max()` or `min()`.
- Count how many numbers are positive, negative, and zero.
- Given a string, count vowels and consonants.
- Create a dictionary called `robot_status` with battery, mode, speed, and temperature. Print each key-value pair.
- Given a list of words, count how many times each word appears using a dictionary.

No LeetCode submission is required. No hash-map problem is required. Only write the drills and notes.

Block 3 - NumPy First Touch: Sensor Arrays

Primary resource: NumPy Quickstart. Read only the beginning parts on `ndarray`, `shape`, `ndim`, `size`, `dtype`, and array creation.

Create `numpy_sensor_arrays.py` with these tasks:

- Create a NumPy array of 10 mock distance sensor readings.
- Print `shape`, `ndim`, `size`, `dtype`, `mean`, `min`, and `max`.
- Create a 2D array representing 5 robot poses: `x`, `y`, `theta`.
- Print the first pose, last pose, and mean `x/y` position.
- Write 5 lines explaining why robotics data is represented as arrays/tensors.

Block 4 - C++ Basics Orientation

Do not solve LeetCode in C++ today. We are only beginning C++ carefully because later ROS 2 and interviews will need it.

Use one source only today: [learncpp.com](https://www.learncpp.com) Chapter 1 or any C++ compiler practice you already have. Focus only on input/output and loops.

Create folder `Day03/cpp_basics/` and write:

- `hello_robot.cpp` - print your target role and one robotics goal.
- `sum_numbers.cpp` - input `n`, then input `n` numbers, print sum and average.
- `cpp_basics_notes.md` - explain `cout`, `cin`, `int`, `double`, `for` loop, `vector` in simple words.

Block 5 - Git Commit and Journal

Commit message: Day 3 revised: fundamentals without LeetCode

Expected folder structure:

```
robot-learning-journey/  
  Day03/  
    countdown_timer.py  
    sensor_threshold_checker.py  
    robot_command_menu.py  
    function_calculator.py
```

```
dsa_foundation_drills.py
numpy_sensor_arrays.py
cpp_basics/
  hello_robot.cpp
  sum_numbers.cpp
  cpp_basics_notes.md
journal.md
```

Today Completion Checklist

- CS50P Loops section studied or revised.
- Python control flow documentation skimmed.
- 4 Python loop/function programs completed.
- DSA foundation drills completed without LeetCode.
- NumPy sensor array program completed.
- 2 very small C++ programs completed.
- Git commit completed.
- Journal written honestly.

Daily Completed Report Template

```
Day 3 completed report:
1. Hours studied:
2. Python files completed:
3. DSA foundation drills completed:
4. NumPy completed:
5. C++ files completed:
6. GitHub commit link:
7. What was easy:
8. What was hard:
9. What I skipped:
10. Questions:
```

Important Rule for the Next Few Days

LeetCode is paused until the basic concepts are comfortable. We will restart only after you understand arrays/lists, strings, dictionaries/hash maps, and simple time complexity. This is not a delay. This is the correct foundation-building path.